Correlation

2026-03-22

```r
# install.packages("corrplot")
library(corrplot)
```

```
## corrplot 0.95 loaded
```

2.loading data

```r
# Input file (use full path, safest way)
inputFile = "/Users/libingchen/Desktop/R_plot/5_Correlation/corelation.txt"

# Read data
rt = read.table(inputFile, sep = "\t", header = TRUE, row.names = 1)

# Transpose (samples × variables)
rt = t(rt)

# Check data
dim(rt)
```

```
## [1] 433  20
```

```r
head(rt[, 1:5])
```

```
##              CCL21    IGFBP6      MMP7 LOC441763      KRT7
## GSM2235556 6.513434  9.631472  9.011758  11.55812  6.081780
## GSM2235557 9.112582 11.186781 12.384619  10.28361 12.318031
## GSM2235558 8.802619  9.866587 13.044805  11.41511  8.086683
## GSM2235559 9.174393  9.673667 12.596247  12.49686 13.365237
## GSM2235560 9.033568 11.204261  9.682779  10.98048  9.914753
## GSM2235561 9.306240 11.104253 12.720841  10.60821 10.941507
```

3.correlation

```r
# Compute Pearson correlation matrix
M = cor(rt, method = "pearson")

# Preview correlation matrix
dim(M)
```

```
## [1] 20 20
```

```r
round(M[1:5, 1:5], 2)
```

```
##           CCL21 IGFBP6  MMP7 LOC441763 KRT7
## CCL21      1.00   0.44  0.13      0.07 0.01
```

```
## IGFBP6     0.44   1.00  0.14      0.29 0.15
## MMP7       0.13   0.14  1.00     -0.07 0.25
## LOC441763  0.07   0.29 -0.07      1.00 0.10
## KRT7       0.01   0.15  0.25      0.10 1.00
```

4.p-value

```
# Function to compute p-values for correlation matrix
cor.mtest <- function(mat, ...) {
  mat <- as.matrix(mat)
  n <- ncol(mat)
  p.mat <- matrix(NA, n, n)
  diag(p.mat) <- 0

  for (i in 1:(n - 1)) {
    for (j in (i + 1):n) {
      tmp <- cor.test(mat[, i], mat[, j], ...)
      p.mat[i, j] <- p.mat[j, i] <- tmp$p.value
    }
  }
  colnames(p.mat) <- rownames(p.mat) <- colnames(mat)
  p.mat
}

# Calculate p-value matrix
p.mat = cor.mtest(rt)

# Preview
dim(p.mat)
```

```
## [1] 20 20
```

```
round(p.mat[1:5, 1:5], 3)
```

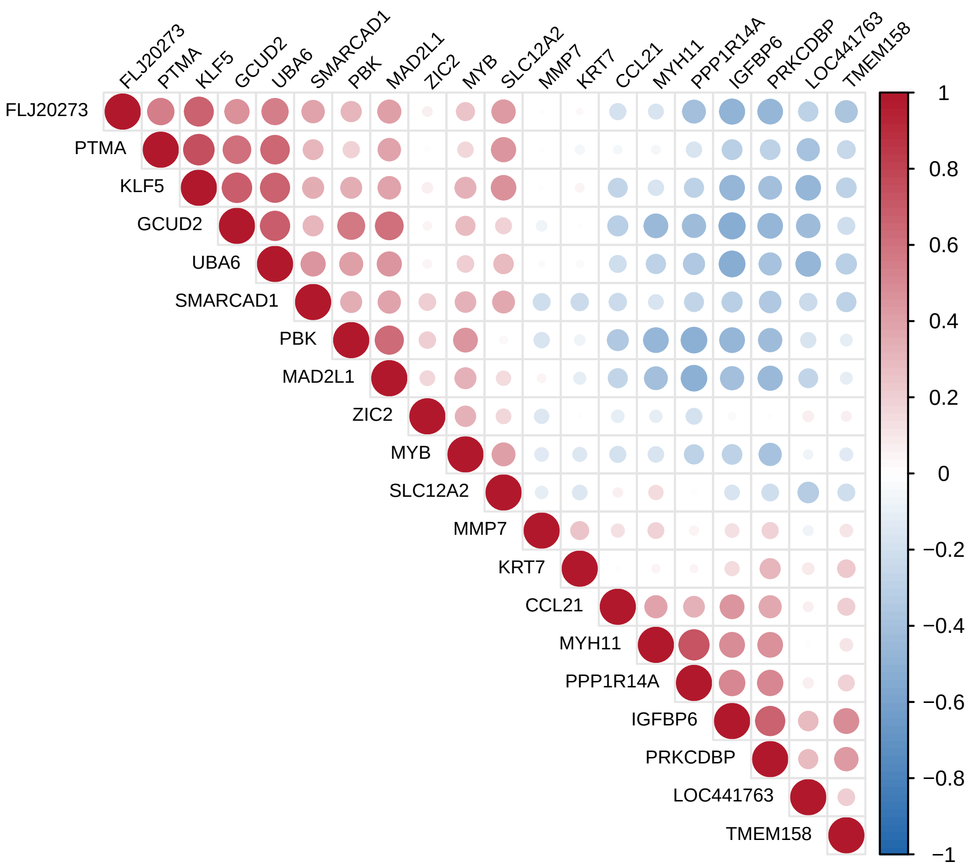
```
##           CCL21 IGFBP6  MMP7 LOC441763  KRT7
## CCL21     0.000  0.000 0.009     0.158 0.791
## IGFBP6    0.000  0.000 0.004     0.000 0.002
## MMP7      0.009  0.004 0.000     0.148 0.000
## LOC441763 0.158  0.000 0.148     0.000 0.045
## KRT7      0.791  0.002 0.000     0.045 0.000
```

```
write.csv(rt,
          file = "/Users/libingchen/Desktop/R_plot/5_Correlation/rt.csv",
          row.names = TRUE)
```

5.plot

```
library(corrplot)

inputFile = "/Users/libingchen/Desktop/R_plot/5_Correlation/corelation.txt"

rt = read.table(inputFile, sep = "\t", header = TRUE, row.names = 1)
rt = t(rt)
```

```r
M = cor(rt, method = "pearson")

corrplot(M,
         method = "circle",
         order = "hclust",
         type = "upper",
         col = colorRampPalette(c("#2166AC", "white", "#B2182B"))(100),
         tl.col = "black",
         tl.cex = 0.6,
         tl.srt = 45,
         cl.cex = 0.7,
         addgrid.col = "grey90"
)
```

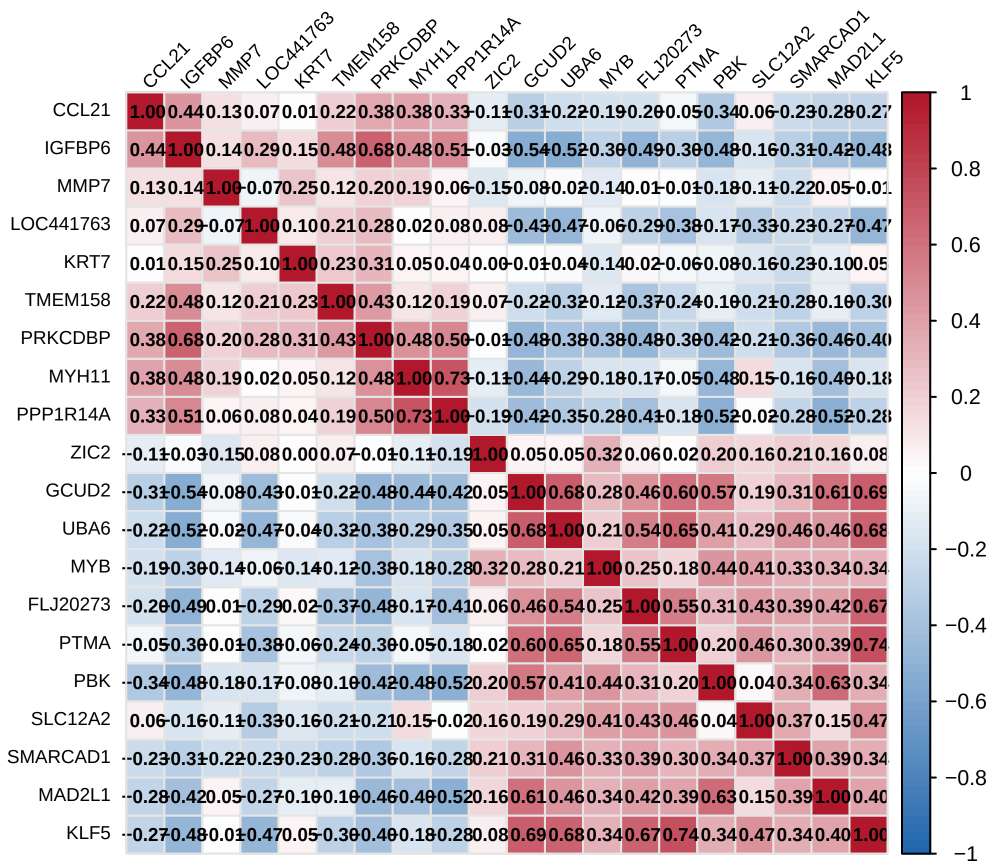



```r
png("/Users/libingchen/Desktop/R_plot/5_Correlation/corplot1.png",
    width = 6, height = 6, units = "in", res = 300)

corrplot(M,
         method = "circle",
         order = "hclust",
         type = "upper",
         col = colorRampPalette(c("#2166AC", "white", "#B2182B"))(100),
         tl.col = "black",
         tl.cex = 0.6,
         tl.srt = 45,
         cl.cex = 0.7,
```

```r
         addgrid.col = "grey90"
)

dev.off()
```

```
## pdf
##   2
```

```r
corrplot(M,
         order = "original",
         method = "color",
         col = colorRampPalette(c("#2166AC", "white", "#B2182B"))(100),
         addCoef.col = "black",
         number.cex = 0.6,
         diag = TRUE,
         tl.col = "black",
         tl.cex = 0.6,
         tl.srt = 45,
         addgrid.col = "grey90",
         cl.cex = 0.7
)
```




```r
png("/Users/libingchen/Desktop/R_plot/5_Correlation/corplot2.png",
    width = 6, height = 6, units = "in", res = 300)

corrplot(M,
         order = "original",
```

```r
         method = "color",
         col = colorRampPalette(c("#2166AC", "white", "#B2182B"))(100),
         addCoef.col = "black",
         number.cex = 0.6,
         diag = TRUE,
         tl.col = "black",
         tl.cex = 0.6,
         tl.srt = 45,
         addgrid.col = "grey90",
         cl.cex = 0.7
)

dev.off()
```

```
## pdf
##   2
```